Question:

At birth, the least developed part of the brain is the

Answer Choices:

A. visual system

B. thalamus

C. frontal lobe

D. brain stem

E. parietal lobe

F. cerebellum

G. cortex

H. basal ganglia

I. hypothalamus

J. limbic system

Reasoning + Answer:

<think>

Neural structures are developmentally graded: the earliest and most fundamental systems mature first, while complex, association-dominated regions take longer to reach full function.

- Core âĢľprimitiveâĢĿ systems that control fundamental survival functions (life-support, arousal, breathing, heart rate) must be operational at birth:
- Brain stem (including medulla, pons, midbrain with reticular formation) regulates basic homeostatic functions and the sleepâĢĵwake cycle, functional even at birth.
- Hypothalamus supports vital autonomic and neuroendocrine homeostasis, present and largely functional at birth.
- In contrast, the cerebral cortex (especially association cortex) undergoes extended maturation, including synaptic pruning, myelination, and functional organization, and is still developing well into childhood and adolescence.

Applying this to the choices:
- Cortex (G): least mature at birth compared with core survival systems.
- Brain stem (D) and hypothalamus (I): most mature at birth.
- Other listed structures (e.g., thalamus, basal ganglia, cerebellum, limbic system, parietal lobe, visual system, frontal lobe) are generally more developed than the cortex at birth but less so than the brain stem and hypothalamus; the cerebellum is relatively immature, but not as âĢľleast developedâĢĿ as the cortex in the classic neonatal scheme.

Therefore, the least developed part of the brain at birth is the cortex.

Correct answer: G. cortex

</think>

<answer>

G. cortex

</answer>